



PM Accelerator

Weather Analysis



Presented By
**Ghada
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1. Executive Summary

This project analyzes the Global Weather Repository dataset containing 137,998 daily weather observations from cities worldwide, spanning May 2024 to April 2026, with the objective of forecasting future temperature trends using machine learning and time series techniques.

The pipeline covered data cleaning, feature engineering, and model building. Key steps included removing physically impossible sensor readings, engineering 13 new features, eliminating data leakage, and applying PCA to compress air quality variables.

Four models were evaluated. XGBoost (tuned) was the best overall model with $R^2=0.90$ and $MAE=2.36^{\circ}C$. For time series forecasting, Prophet achieved the best results with $MAE=1.75^{\circ}C$ for Cairo, with negative R^2 explained by an anomalous $5^{\circ}C$ temperature drop in the test period compared to the same period the prior year.





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2. Dataset Overview

The Global Weather Repository dataset was sourced from Kaggle, containing daily weather snapshots from cities worldwide spanning May 2024 to April 2026, with no missing values or duplicate records found.

137,995 Total Records

41 Features

Meteorological	Air Quality	Astronomical	Location
Temperature, Wind, Pressure, Precipitation, Humidity, Visibility	CO, Ozone, NO2, SO2, PM2.5, PM10	Sunrise, Sunset, Moonrise, Moon Phase	Country, City, Latitude, Longitude





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3. Data Cleaning & Preprocessing

The dataset was largely clean with no missing values or duplicate records found. Preprocessing focused on identifying and handling physically impossible sensor readings, capping extreme pollution spikes, and normalizing features for modeling.

5 rows

Removed

20 cols.

Modified

137,995 × 41

Final Dataset





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Step	Column(s)	Action	Result
Missing Values	All columns	No action needed	Clean
Duplicates	All columns	No action needed	Clean
Temperature Outliers	temperature_celsius	Removed values > 60°C	3 rows removed
Pressure Outliers	pressure_mb	Removed values > 1084 mb	2 rows removed
Wind Outliers	wind_mph, gust_mph	Capped at 99th percentile	Capped
Precipitation Outliers	precip_mm	Capped at 99th percentile	Capped
Air Quality Outliers	CO, Ozone, NO2, SO2, PM2.5, PM10	Capped at 99th percentile	6 columns capped
Weather Extremes	wind, humidity, UV, visibility	Kept unchanged	Physically valid
Scaling – Weather	13 weather/AQ features	RobustScaler	13 columns scaled
Scaling – Bounded	humidity, cloud, UV, moon	MinMaxScaler	4 columns scaled
Scaling – Precipitation	precip_mm	MinMaxScaler	Re-scaled





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RobustScaler

Used for weather & air quality features with legitimate extreme values. Scales using median & IQR so outliers don't distort the scale.

MinMaxScaler

Used for naturally bounded columns (humidity, cloud, moon illumination) and precipitation. Squeezes values into 0–1 range.



2,382 temperature outliers were detected via IQR — 96% were legitimate cold-climate readings from Mongolia, Canada and Kazakhstan. Only 3 physically impossible values above 60°C were removed."



A precipitation value of 2,112mm was identified as a data entry error — likely a decimal shift from 2.12mm. Capped at the 99th percentile of 2.12mm.





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4. Exploratory Data Analysis

Exploratory analysis revealed distinct global weather patterns, significant air quality disparities across regions, and clear seasonal temperature cycles. Seven key visualizations were produced to uncover the dataset's structure and inform feature engineering decisions.

-29.8°C to 49.2°C

Global Temp Range

33.2% of records

Rainy Readings

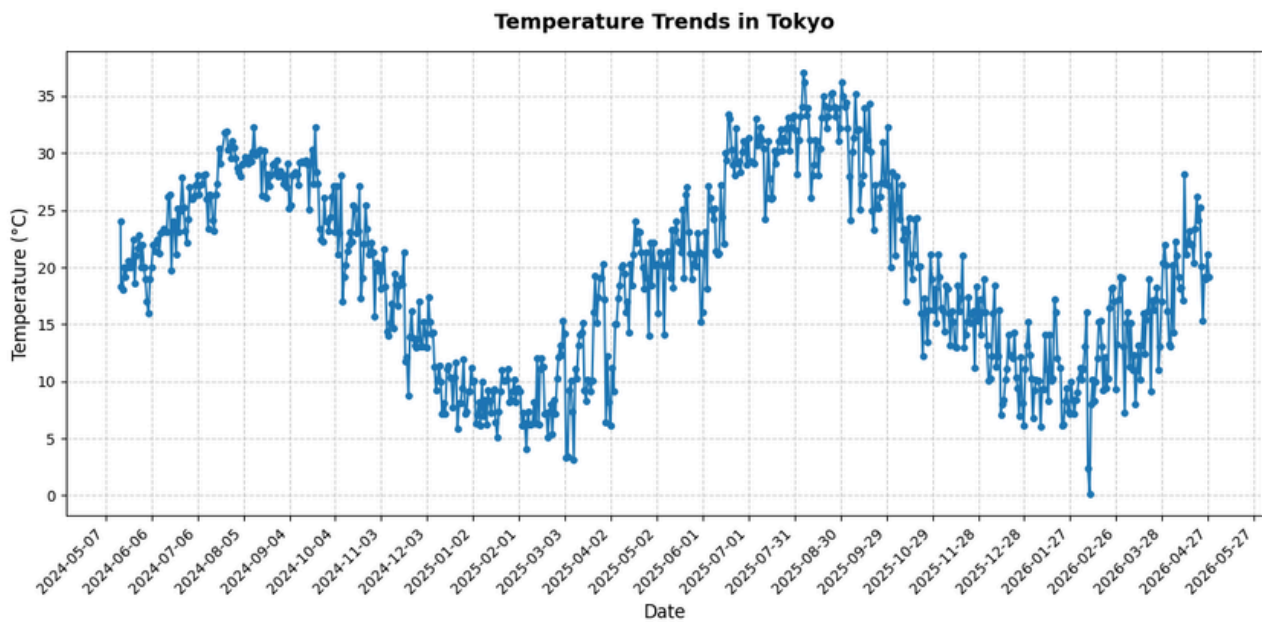
China (109 $\mu\text{g}/\text{m}^3$)

Most Polluted





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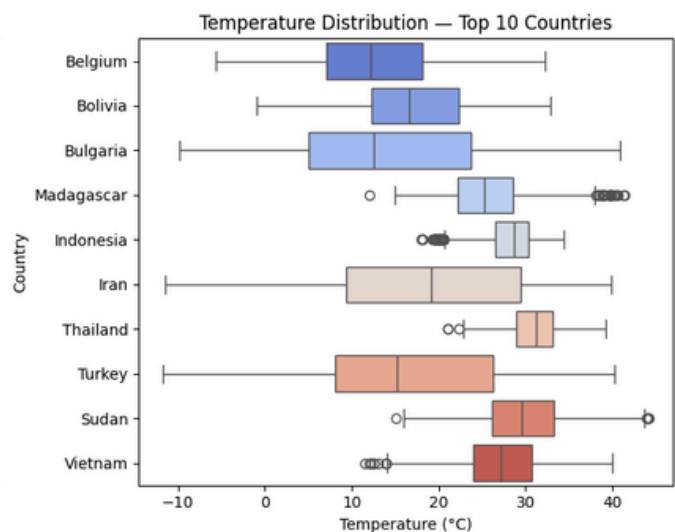
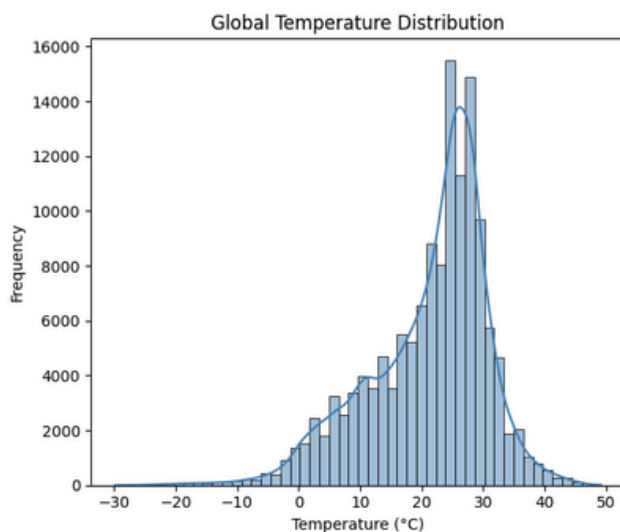
The data exhibits a classic, well-defined annual seasonal pattern. You can clearly identify two distinct seasonal cycles:

- **Summer Peaks:** Temperatures consistently crest between 35°C and 37°C around August of both 2024 and 2025.
- **Winter Troughs:** Temperatures drop significantly to their lowest points between 0°C and 5°C around January/February of 2025 and 2026.





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Global Temperature Distribution:

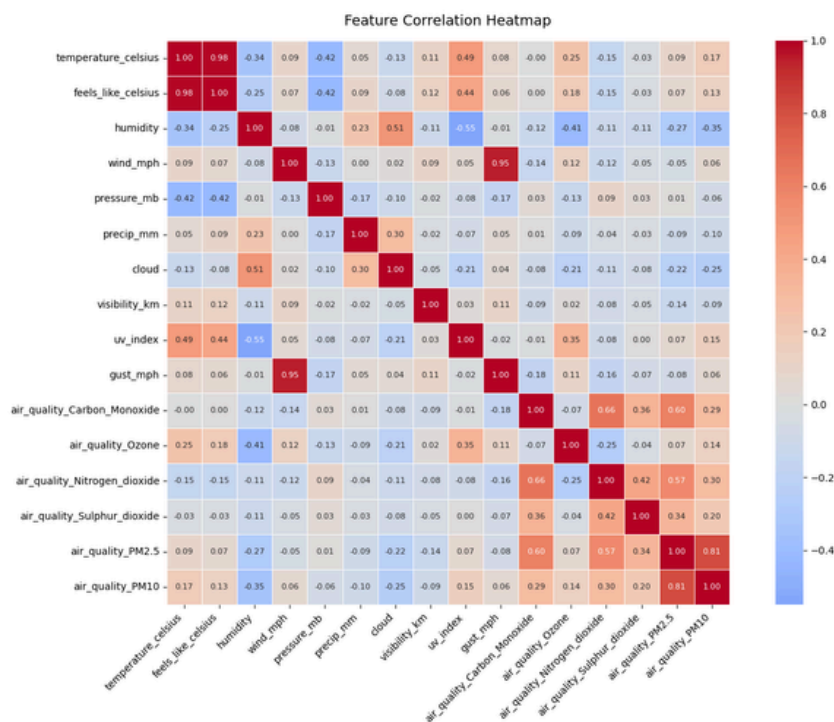
- Slightly left-skewed, with most readings between 10°C and 35°C, indicating a dominance of temperate to tropical cities.
- Peak temperatures are around 25–28°C, reflecting warm-climate cities.
- Cold-climate outliers (e.g., Mongolia, Canada, Kazakhstan) are a minority.

Temperature by Country:

- Belgium & Bulgaria have the widest IQR, showing **high seasonal variability typical of continental Europe**.
- Indonesia & Thailand exhibit tight, high boxes, indicating consistent heat and humidity.
- Sudan & Vietnam have **medians above 28°C**, placing them at the warmest end.
- Iran & Turkey show wide ranges due to extreme seasonal variations.
- Madagascar has outliers reflecting its diverse geography.
- Bolivia has a cool median (~15°C) due to its high-altitude location.



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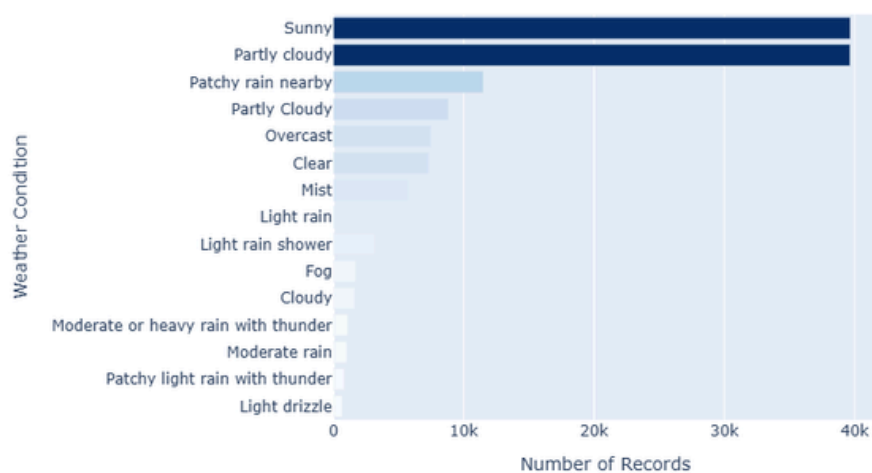


1. The dataset contains several highly correlated variable pairs, such as temperature_celsius with feels_like_celsius (**0.98**) and wind_mph with gust_mph (**0.95**). These strong relationships indicate feature redundancy and potential multicollinearity in predictive models.
2. **Air pollution variables** form a clear correlation cluster. Carbon monoxide, nitrogen dioxide, and particulate matter are strongly related, suggesting common sources such as traffic and industrial emissions.
3. **Ozone levels** are positively associated with **UV index** and negatively associated with humidity, reflecting the role of sunlight in ozone formation and the suppressing effect of humid, cloudy conditions.
4. Variables such as **visibility_km** and **precip_mm** show weak correlations with most other features, indicating they provide relatively independent information that may improve machine learning models.



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Top 15 Most Common Weather Conditions

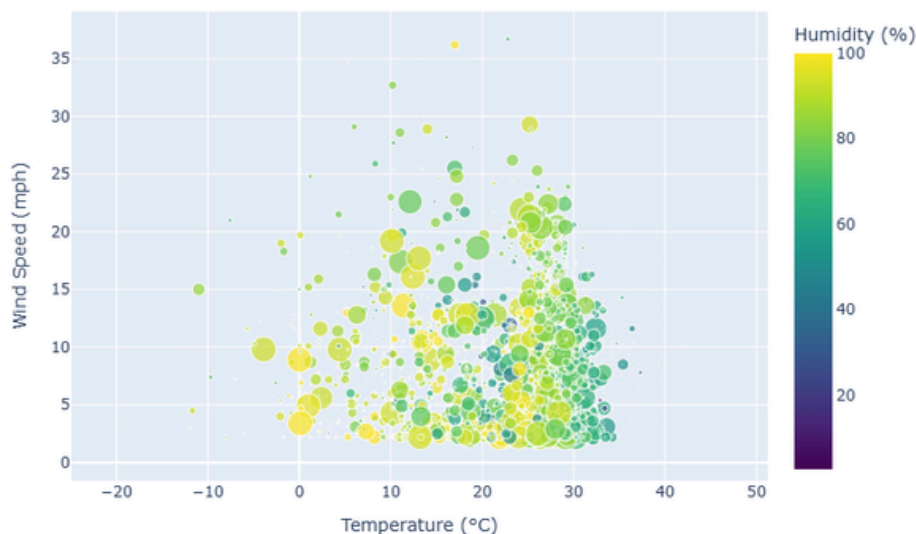


1. The dataset shows strong class imbalance, with **“Sunny”** and **“Partly cloudy”** dominating the observations. Together, these conditions account for most records, meaning predictive models may become biased toward clear-weather predictions.
2. The target labels contain data quality issues caused by inconsistent naming conventions. For example:
 - **“Partly cloudy”** and **“partly Cloudy”** appear as separate categories despite representing the same condition.
 - Similar overlap exists between **“Sunny”** and **“Clear.”**
 - This indicates the need for text normalization and category merging before analysis.
3. Weather events related to **rain** and low visibility form a long-tail distribution with relatively small sample sizes. Rain conditions are fragmented across many subcategories such as light rain, drizzle, and thunder-related events, while conditions like fog occur infrequently.
4. For machine learning tasks, consolidating similar weather categories and addressing class imbalance would improve statistical reliability and model performance.



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Wind Speed vs Temperature (colored by Humidity)

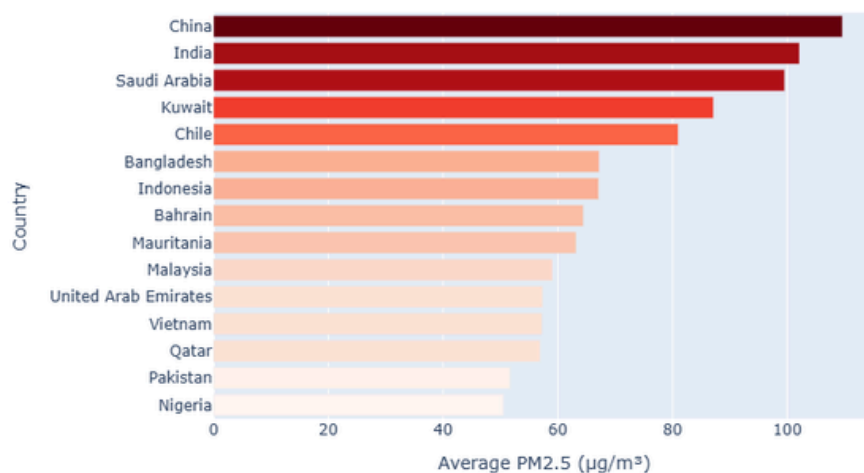


1. The relationship between **temperature** and **wind speed** is non-linear and bounded. Wind speed variability is highest between 15°C and 30°C, while extreme high temperatures above 35°C are associated almost exclusively with low wind speeds, indicating hot and stagnant atmospheric conditions.
2. **Humidity** shows a strong dependence on temperature. Cooler and moderate **temperatures** are dominated by high-humidity conditions, whereas hotter temperatures between 30°C and 40°C correspond to noticeably lower humidity levels, revealing a transition toward dry atmospheric environments.
3. Higher **wind speeds** are generally associated with higher **humidity** levels, while low-humidity observations are concentrated in low-wind regions. This suggests that dry conditions in the dataset tend to occur alongside relatively stagnant air masses.
4. Overall, the scatter distribution highlights **three distinct environmental regimes**:
 - a. Cool and humid conditions at lower temperatures
 - b. Highly variable and dynamic weather between 15°C and 30°C
 - c. Hot, dry, and low-wind conditions at extreme temperatures



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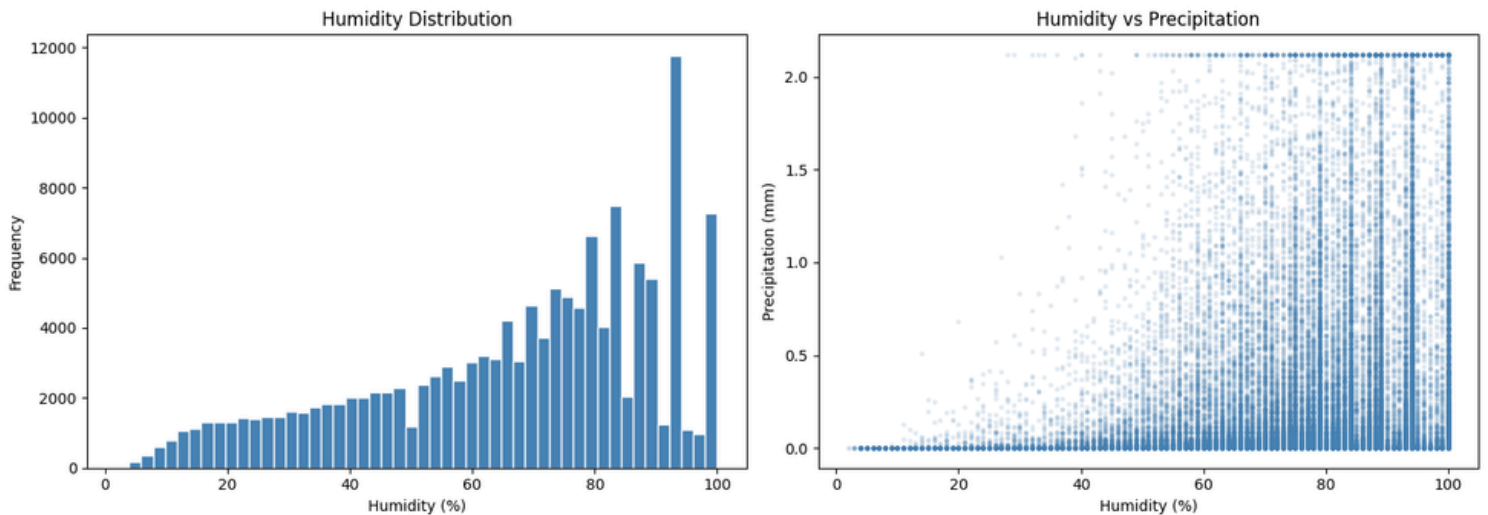
Top 15 Countries by Average PM2.5 (min. 100 records)



1. The visualization identifies a critical high-pollution tier led by **China, India, and Saudi Arabia**, where average PM2.5 concentrations exceed **100 µg/m³**. These values represent extremely hazardous air quality levels and indicate widespread exposure to harmful particulate pollution.
2. The countries in the distribution reflect two major pollution drivers:
 - a. **Industrial and urban emissions** in countries such as China, India, Bangladesh, and Indonesia
 - b. **Natural desert dust and arid climate conditions** in countries such as Saudi Arabia, Kuwait, Qatar, and the UAE
3. After the top-ranked countries, PM2.5 levels gradually decline but remain consistently high, with most nations still exceeding **50 µg/m³**. This demonstrates that severe air pollution is a widespread global issue affecting countries across different regions and economic conditions.
4. Overall, the distribution highlights the combined impact of industrialization, urbanization, geographic conditions, and climate on global particulate pollution patterns.



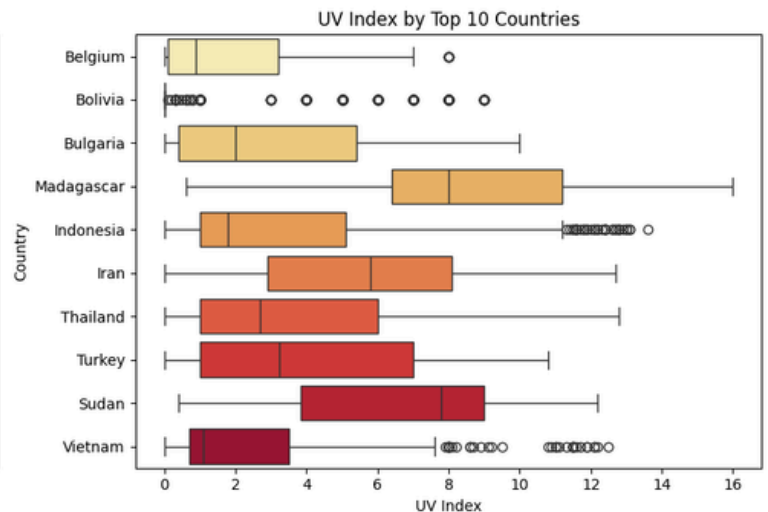
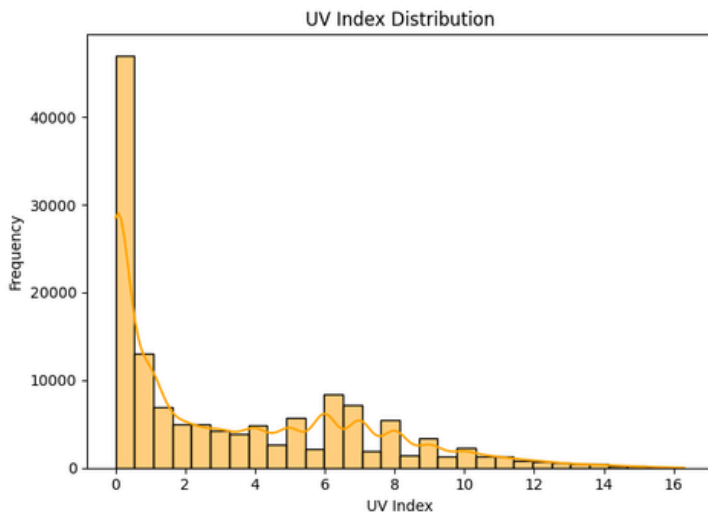
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1. The humidity distribution is **strongly left-skewed**, with most observations concentrated between 90% and 100% humidity. Sharp spikes near specific values, particularly around 93% and 100%, suggest possible sensor rounding or data collection artifacts.
2. The relationship between humidity and precipitation follows a **clear threshold pattern**. Very low humidity levels are associated with almost no rainfall, while precipitation variability increases significantly as humidity rises. This indicates that high humidity is necessary for rainfall but does not guarantee it.
3. The precipitation data shows a visible ceiling effect at **2.1 mm**, where many observations align along a flat upper boundary. This likely reflects data clipping, sensor measurement limits, or preprocessing constraints.
4. From a modeling perspective, the dataset presents challenges including skewed humidity distributions, increasing variance in precipitation at higher humidity levels, and capped precipitation values, all of which may affect regression model performance.



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1. The global UV Index distribution is strongly **right-skewed**, with most observations concentrated between 0 and 2, representing **low or negligible UV exposure**. A secondary concentration between 6 and 8 reflects typical **midday sunlight conditions**, while extreme UV values occur less frequently.
2. The boxplot reveals clear geographic variation in UV exposure. Countries such as **Sudan and Madagascar** show consistently high UV levels with broad distributions, whereas countries like **Belgium and Bulgaria** exhibit lower and more compact UV ranges due to weaker solar intensity.
3. Some countries display unusual distribution patterns. **Bolivia** shows a compressed low-UV distribution with several high outliers, while **Vietnam** exhibits mostly low UV levels with intermittent spikes, likely reflecting seasonal or climatic variability.
4. From a modeling perspective, the UV Index feature is **highly skewed and zero-inflated**, which may reduce the effectiveness of standard normalization techniques. Feature transformations or categorical UV risk groupings may provide more stable inputs for machine learning models.



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5. Climate Analysis

Climate analysis examined long-term temperature trends, seasonal patterns, and year-over-year variations across the dataset's nearly two-year span from May 2024 to April 2026.

-0.325°C/month

Monthly Trend

July'24 (26.80°C)

Hottest Month

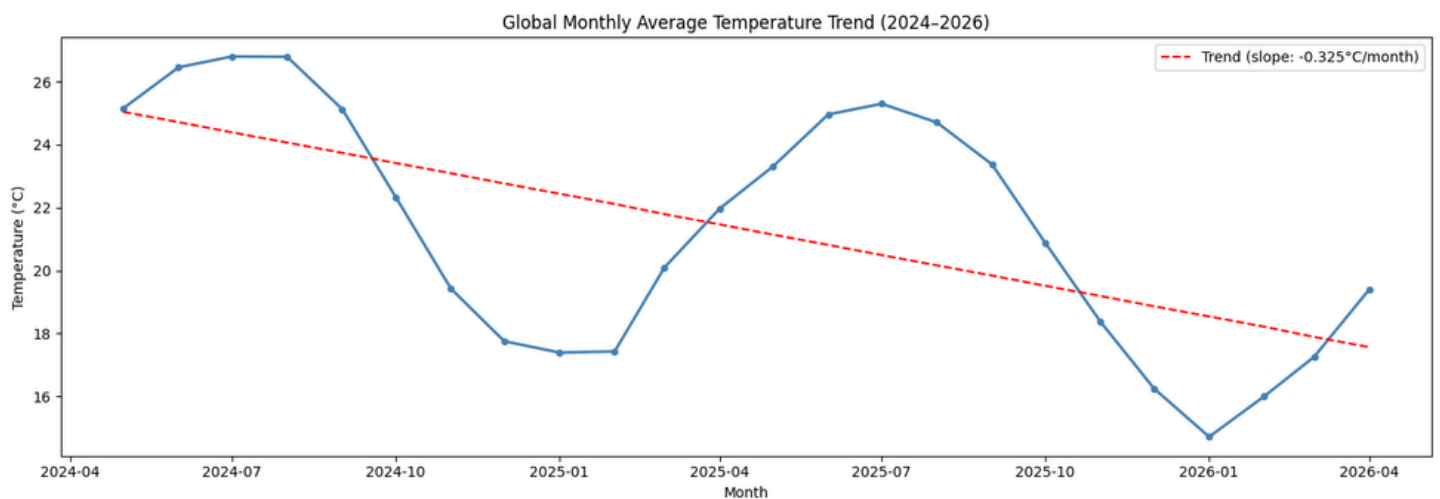
January'26 (14.71°C)

Coldest Month





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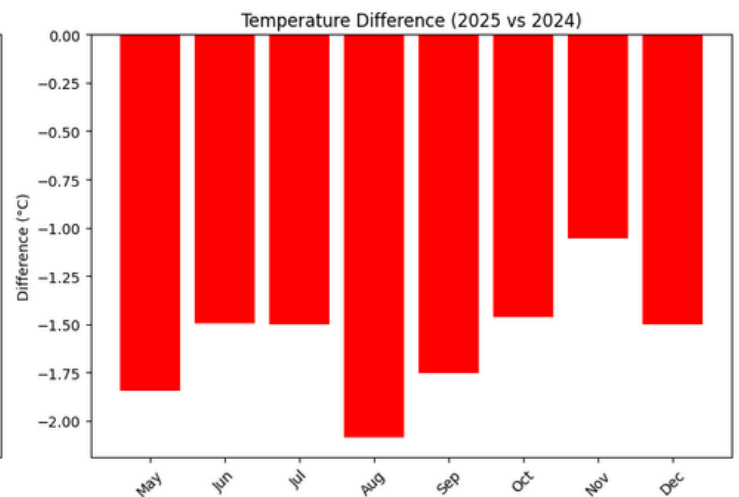
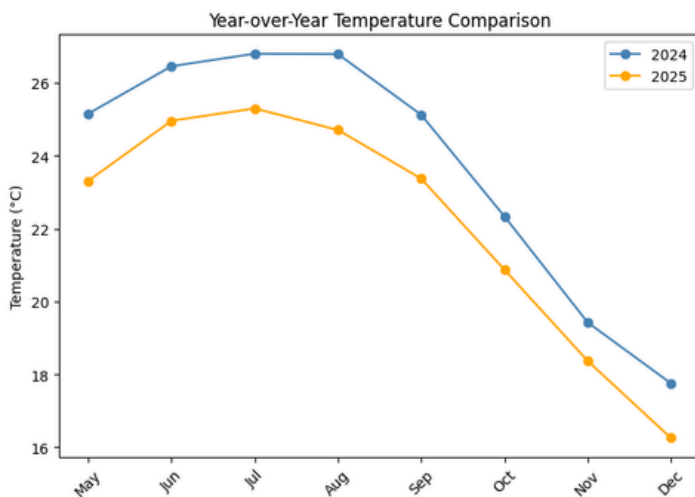


Two complete seasonal cycles are clearly visible across the observation period, with summer peaks around 26–27°C and winter troughs around 17°C. The overall trend line shows a -0.325°C per month slope, however this is largely a statistical artifact of the dataset starting in May 2024 (summer peak) and ending in April 2026 (spring), rather than a genuine long-term cooling signal.

Season	Avg Temp (°C)	Min Temp (°C)	Max Temp (°C)	Std Dev	Key Insight
Summer	25.82	-3.7	49.2	7.09	Warmest season with extreme heat events
Fall	21.58	-20.1	47.2	8.38	Moderate temperatures with noticeable variability
Spring	20.92	-23.9	47.7	9.00	Transitional season with fluctuating conditions
Winter	16.60	-29.8	41.4	11.42	Coolest and most variable season



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2025 was consistently cooler than 2024 across all common months, with every month showing a negative difference. The largest gap occurred in August, where 2025 was approximately 2°C cooler than 2024. Overall 2025 averaged 21.17°C compared to 2024's 23.73°C – a significant 2.56°C year-over-year cooling. The anomalously cold January 2026 (14.71°C vs 17.39°C in January 2025) further extends this cooling pattern into 2026.





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6. Geographical Patterns

Geographical analysis revealed stark climate contrasts across regions, driven by latitude, proximity to oceans, altitude, and land mass size. 186 countries with sufficient records were analyzed across temperature, humidity, and seasonal variation dimensions

186
Countries

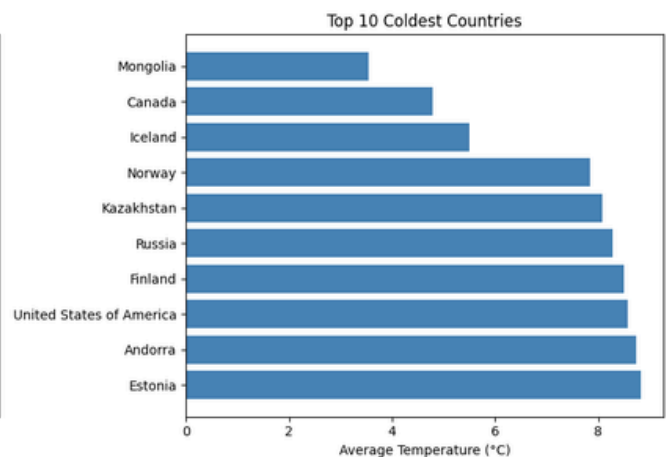
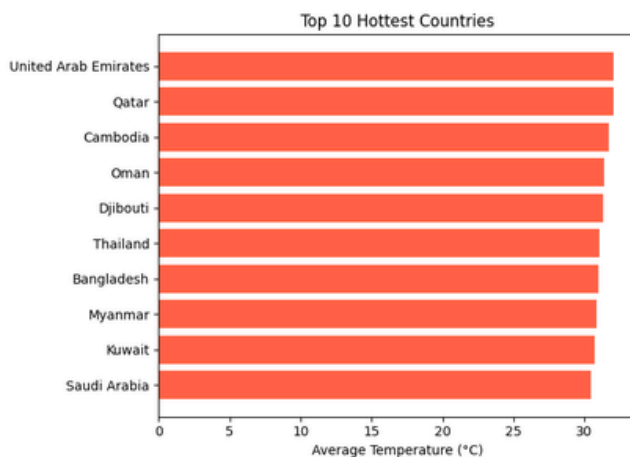
UAE (32.05°C)
Hottest Country

Mongolia (3.55°C)
Coldest Country





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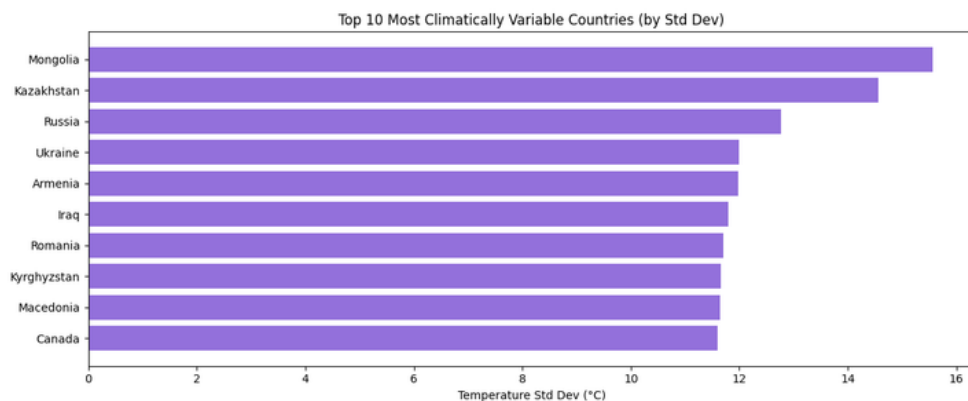
Top 10 Hottest: UAE and Qatar lead at **~32°C**, dominated by desert heat. Southeast Asian countries (Cambodia, Thailand, Myanmar, Bangladesh) follow with consistently high but stable temperatures year-round. Kuwait and Saudi Arabia show the highest variability among hot countries, experiencing scorching summers of **49°C** yet surprisingly cold winters around **6–7°C**.

Top 10 Coldest: Mongolia is the coldest country (**3.55°C**) with the widest temperature range in the dataset (**-29.8°C to 34.3°C**). Iceland despite being cold shows the lowest std (**4.61°C**) among cold countries, stabilized by the Gulf Stream. Canada and Russia follow with harsh continental winters and moderate summers.



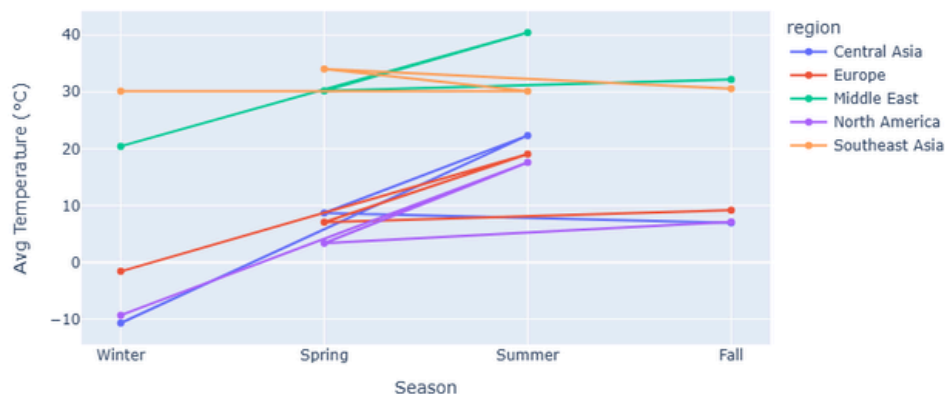


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Most Variable Countries: Mongolia and Kazakhstan lead variability with std exceeding 14°C. Iraq uniquely appears in both the hottest and most variable lists – summers reach 49.1°C while winters drop to 2°C, a 47°C annual swing. Eastern European countries (Romania, Ukraine, Macedonia) sit on the boundary between oceanic and continental climates, producing high variability.

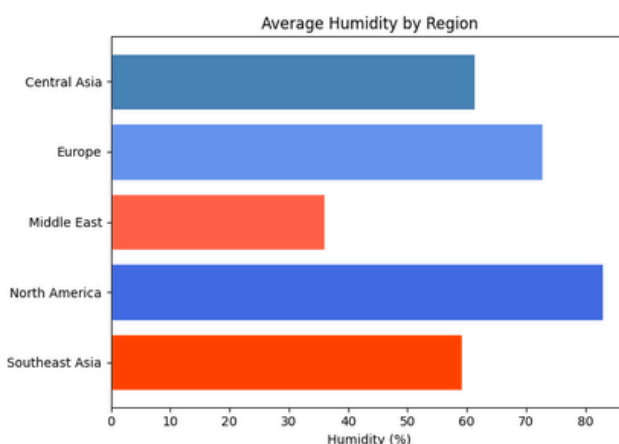
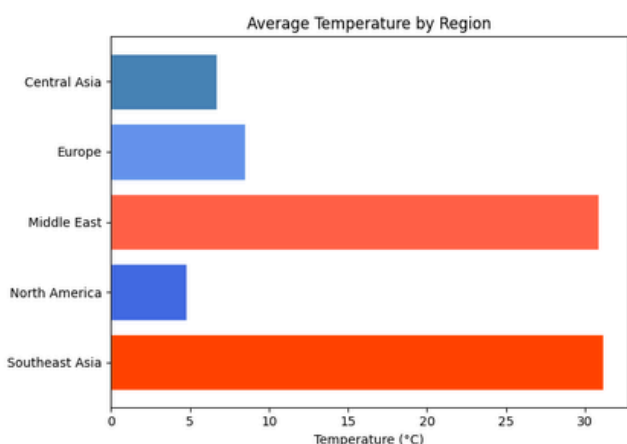
Seasonal Temperature Patterns by Region



The Middle East shows the most dramatic seasonal swing, jumping from 21°C in winter to 40°C in summer. Southeast Asia is the flattest line, barely moving between seasons confirming year-round tropical stability. All regions converge in Fall, making it the most climatically similar season globally.



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Middle East and Southeast Asia share nearly identical average temperatures (~31°C) but are climatically opposite – the Middle East has only 36% humidity (dry desert heat) while Southeast Asia has 59% humidity (tropical wet heat). North America is the coldest region yet the most humid (83%), while an inverse relationship between temperature and humidity is confirmed across all regions.

Regions temperature analysis results

Region	Insight
Middle East	Hottest region on average
Southeast Asia	Warm and most temperature-stable region
Central Asia	Largest temperature variation
North America	Coollest average temperatures



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Regions humidity analysis results

Region	Insight
North America	Highest average humidity
Middle East	Lowest average humidity
Europe	Consistently humid conditions
Central Asia	Most variable humidity levels
All Regions	Humidity occasionally reached 100%

Overall Climate Patterns

Observation	Insight
Middle East	Hot and dry climate
Southeast Asia	Stable tropical climate
Central Asia	Most extreme seasonal variability
Europe	Temperate weather conditions



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7. Feature engineering

Feature engineering was performed in two phases — creating new meaningful features from existing data, and resolving structural issues identified during EDA including multicollinearity, data leakage, and redundant unit duplicates

13

new features

12 columns

dropped

25

final feature set





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Feature	Description
season	Winter / Spring / Summer / Fall extracted from month
temp_range	Difference between actual and feels-like temperature
heat_index	Combined temp + humidity indicator (valid when temp > 27°C & humidity > 40%)
wind_chill	Perceived cold temperature with wind (valid when temp < 10°C & wind > 3mph)
is_rainy	Binary flag — 1 if precipitation > 0
air_quality_index	Normalized average of all 6 pollutants
time_of_day	Morning / Afternoon / Evening / Night from hour
is_daytime	Binary flag derived from sunrise and sunset hours
condition_grouped	48 weather conditions collapsed into 9 meaningful categories
pollution_PC1/2/3	6 air quality columns compressed into 3 PCA components
uv_category	UV index binned into WHO ordinal tiers (Low → Extreme)
hour_sin/cos	Cyclical encoding of hour to capture daily periodicity
month_sin/cos	Cyclical encoding of month to capture seasonal periodicity



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Data Leakage Discovery

During initial modeling, Random Forest achieved a perfect $R^2=1.0$ with only two features dominating — `heat_index` (50.5%) and `wind_chill` (49.5%). Both features were derived directly from `temperature_celsius`, the target variable, making them perfect predictors by construction. All three leaking features (`heat_index`, `wind_chill`, `temp_range`) were removed before modeling.

Additional Structural Fixes

Nine unit-duplicate columns were dropped (e.g. `temperature_fahrenheit`, `wind_kph`, `pressure_in`) as they carried identical information to their metric counterparts. Six air quality columns were compressed into 3 PCA components capturing 82.5% of the original variance, reducing dimensionality without significant information loss.





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8. Model Building & Evaluation

Four models were built to forecast temperature: Linear Regression as a baseline, Random Forest, XGBoost with hyperparameter tuning, and Prophet for single-city time series forecasting. A time-based 80/20 train/test split was used to preserve temporal order and reduce future-data leakage. After the first baseline check, target-derived variables (`heat\_index`, `wind\_chill`, and `temp\_range`) were removed, and the final model results below use the leakage-corrected feature set.

Regression Preprocessing and Standardization

Linear regression preprocessing was performed with strict leakage prevention. Scaling was applied only after the train/test split, ensuring each scaler was fitted on the training set and then applied to both training and testing data. After removing leakage-related variables, the final regression dataset contained 25 predictors.





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Feature scaling was applied using different methods based on variable type:

- Skewed continuous variables (**wind_mph**, **pressure_mb**, **visibility_km**, **pollution_PC1**, **pollution_PC2**, **pollution_PC3**) were transformed using **RobustScaler** to reduce outlier influence.
- Bounded percentage-style variables (**humidity**, **cloud**, **moon_illumination**) were scaled using **MinMaxScaler** to preserve their 0–1 structure.
- All remaining numeric features, including location and time-based variables, were standardized using **StandardScaler** for consistent mean-variance scaling.

Tree-based models (Random Forest and XGBoost) were trained using the same cleaned feature set but without any scaling. This is because these models are not sensitive to feature magnitude and can naturally handle raw numeric inputs.

Model 1 — Linear Regression (Baseline)

Linear Regression was applied as the baseline model using scaled numerical features. After removing the target-derived leakage variables, the model achieved an R2 of **0.45** and MAE of **6.37** deg C. This confirmed that temperature prediction is highly non-linear and cannot be adequately captured by a linear model alone.





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Model 2 — Random Forest

Random Forest significantly improved over the baseline, achieving $R^2 = 0.86$ and MAE = **2.86** deg C on the leakage-corrected feature set. Feature importance analysis identified latitude as the dominant predictor at **38.3%**, followed by atmospheric pressure at **18.4%**, month cosine seasonality at **12.4%**, humidity at **10.1%**, and day-of-year at **6.3%**. These drivers align well with meteorological theory: temperature is strongly shaped by geography, atmospheric conditions, and seasonal cycles.

Model 3 — XGBoost With Hyperparameter Tuning

XGBoost improved further over Random Forest. The default XGBoost model reached MAE = **2.52** deg C and RMSE = **3.69** deg C, and the tuned version achieved the best overall regression performance with $R^2 = 0.90$, MAE = **2.36** deg C, and RMSE = **3.45** deg C. Hyperparameter tuning used GridSearchCV with TimeSeriesSplit across 540 parameter combinations.

Parameter	Best Value
n_estimators	300
max_depth	6
learning_rate	0.1
subsample	1.0





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Model 4 — Ensemble Models

Three ensemble approaches were tested using the two strongest tree-based models, Random Forest and tuned XGBoost. The goal was to check whether combining model predictions could improve generalization beyond the best single model.

Ensemble Method	Description	MAE (deg C)	RMSE (deg C)	R2
Simple average	Average of Random Forest and tuned XGBoost predictions	2.49	3.66	0.8893
Weighted average	60% tuned XGBoost, 40% Random Forest	2.44	3.59	0.8932
Stacking	Random Forest and tuned XGBoost predictions passed into a Linear Regression meta-model	2.54	3.71	0.8859

The weighted ensemble was the best ensemble, but it did not outperform tuned XGBoost. This suggests that XGBoost was already capturing most of the useful non-linear signal, while Random Forest added some diversity but not enough to improve the final prediction. The stacking ensemble also underperformed because the meta-learner could not find a better combination than directly weighting the stronger XGBoost model more heavily.



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Model 5 — Prophet Time Series Forecasting

Prophet was applied to forecast **Cairo's daily average temperature** as a representative single-city time series. After grid search tuning, the best configuration used **changepoint_prior_scale = 0.01** and **seasonality_prior_scale = 1.0**, achieving MAE = **1.75** deg C over a **30-day** horizon. The negative R2 of **-0.09** indicates that, although average error was low, the model did not explain variance better than a simple mean-based benchmark during the test window. This was attributed to an anomalous climate shift: April 2026 temperatures in Cairo were around 5 deg C colder than the same period in 2025, making accurate forecasting from historical seasonality alone difficult.

Model	MAE (deg C)	RMSE (deg C)	R2	Notes
Linear Regression	6.37	8.18	0.45	Leakage-corrected baseline
Random Forest	2.86	4.15	0.86	Strong non-linear learner
XGBoost default	2.52	3.69	0.89	Better than Random Forest
XGBoost tuned	2.36	3.45	0.90	Best overall regression model
Ensemble model (weighted)	2.44	3.59	0.8932	Hybrid learner combining Random Forest and XGBoost outputs
Prophet tuned	1.75	2.11	-0.09	Best single-city short-term time series model



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9. Conclusions & Recommendations

This project successfully demonstrated a full data science pipeline applied to global weather forecasting, from raw data cleaning through advanced modeling and climate analysis. Several key findings emerged across preprocessing, modeling, and analytical stages.

XGBoost (tuned)

Best Model

90.1%

Best R^2

2.36°C

Best MAE





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Limitations

- Dataset captures only ~190 cities, not full global coverage
- Time series limited to 2 years — insufficient for long-term climate trend conclusions
- Prophet's negative R^2 during anomalous climate periods highlights sensitivity to unseen events
- Regional analysis limited to selected countries, not full continent representation



Future Recommendations

- Incorporate real-time weather API feeds for live forecasting
- Extend dataset to 10+ years for robust climate trend detection
- Add LSTM or Transformer-based deep learning models for time series
- Include El Niño/La Niña indices as external regressors in Prophet
- Expand spatial analysis with full continent-level coverage

